XIAOFENG TAN

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RESEARCH INTERESTS

RL, AIGC, and world models, with a focus on human-centric perception, generation, and understanding.

EDUCATION

Southeast University (SEU), Nanjing, China — Sept 2024 – Present
CSRankings (Worldwide, 2024–2026): AI #23 · CV #57 · ARWU 2026 #89
M.Sc. in Computer Science — PALM Lab, supervised by Prof. Hongsong Wang — *GPA:* **88.6**/100
First-Class Academic Scholarship (top 10%) & Tencent-SEU Scholarship (top 2.5%); Summer Camp Rank 2

Shenzhen University (SZU), Shenzhen, China — Sept 2020 – Jun 2024
CSRankings (Worldwide, 2024–2026): AI #27 · CV #32 · ARWU 2026 151–200
B.S. in Computer Science & B.S. in Mathematics *(Dual Degree, Honor)* — *GPA:* **88.9**/100 · **4.07**/4.50
Outstanding Graduate Representative (1/300+) and graduation speaker; featured in SZU's 2024 Admissions Guide

SELECTED PUBLICATIONS

* indicates equal contribution

♦ **When Policy Entropy Constraint Fails: Preserving Diversity in Flow-based RLHF via Perceptual Entropy**
Xiaofeng Tan, Jun Liu, Bin-Bin Gao, Yuanting Fan, Xi Jiang, Chengjie Wang, Hongsong Wang, Feng Zheng
NeurIPS'26 · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A*)
Spotlight (292/30,709 submissions, 0.95%)
[arXiv] [Project]

♦ **MotionRFT: Unified Reinforcement Fine-Tuning for Text-to-Motion Generation**
Xiaofeng Tan*, Wanjiang Weng*, Hongsong Wang, Fang Zhao, Xin Geng, Liang Wang
TPAMI'26 · *IEEE Transactions on Pattern Analysis and Machine Intelligence* (CCF-A, SCI-Q1)
Extended from EasyTune (ICLR'26).
[arXiv] [Project] [Code]

EasyTune: Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation
Xiaofeng Tan*, Wanjiang Weng*, Haodong Lei, Hongsong Wang
ICLR'26 · *International Conference on Learning Representations* (CCF-A, CORE-A*)
[PDF] [Project] [Openreview]

ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment
Wanjiang Weng*, Xiaofeng Tan*, Junbo Wang, Guo-Sen Xie, Pan Zhou, Hongsong Wang
AAAI'26 · *AAAI Conference on Artificial Intelligence* (CCF-A, CORE-A*)
[PDF] [Project] [Code]

FG-Diff: Frequency-Guided Diffusion Model with Perturbation Training for Skeleton-Based VAD
Xiaofeng Tan, Hongsong Wang, Xin Geng, Liang Wang
TIP'26 · *IEEE Transactions on Image Processing* (CCF-A, SCI-Q1)
[Project] [PDF] [Code]

SoPo: Text-to-Motion Generation Using Semi-Online Preference Optimization
Xiaofeng Tan, Hongsong Wang, Xin Geng, Pan Zhou
NeurIPS'25 · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A*)
[Project] [PDF] [Code]

Fuzzy Granule Density-Based Outlier Detection with Multi-Scale Granular Balls
Can Gao (Supervisor), Xiaofeng Tan, Jie Zhou, Weiping Ding, Witold Pedrycz
TKDE'25 · *IEEE Transactions on Knowledge and Data Engineering* (CCF-A, SCI-Q1)
[PDF] [Code]

Dual-Grained Policy Optimization for Hallucination Reduction in Flow-based RL
Xiaofeng Tan, Jun Liu, Yuanting Fan, Bin-Bin Gao, Xi Jiang, Xiaochen Chen, Jinlong Peng, Chengjie Wang, Hongsong Wang, Feng Zheng
Under Review
[Project] [arXiv] [Code]

SERVICE

Conference Reviewer: NeurIPS 2026 (**Top Reviewer**), ICML 2026 (**Gold Reviewer**), AAAI 2027, ICLR 2027
Journal Reviewer: Transactions on Machine Learning Research (TMLR), International Journal of Data Science and Analytics (IJDSA)

RESEARCH EXPERIENCE

Tencent Inc., Shenzhen, China — Mar 2026 – Present
Research Intern @ HY — *Mentor:* Jiaxin Lin
• Spatial reward modeling and curriculum-guided RL for text-to-image generation: tool-grounded spatial verification with SpatialCritic and rollouts progressing from foundational skills to complex compositions (Getting Space Right, under review; joint work with UC Merced).

University of California, Merced, Merced, USA (Remote) — Apr 2026 – Present
Remote Visiting Student @ VLLab — *Mentor:* Ming-Hsuan Yang
• Remote collaboration on spatial reasoning and reward reliability for image generation, through the joint Getting Space Right project with Tencent HY.

Tencent Inc., Shenzhen, China — Jul 2025 – Mar 2026
Research Intern @ YouTu Lab — *Mentor:* Jun Liu
• Analyzed why policy entropy can remain constant while perceptual diversity collapses in flow-based RLHF; developed perceptual-entropy regularization to improve the quality–diversity trade-off (PEC, NeurIPS'26 Spotlight).
• Investigated visual hallucinations caused by limited exploration and trajectory imitation; developed dynamic-granularity rollouts and consistent policy-gradient optimization (Consistent-RFT, under review).

Singapore Management University, Singapore (Remote) — Jul 2024 – Mar 2025
Remote Visiting Student @ Language and Vision Lab (LV Lab) — *Mentor:* Pan Zhou
• Developed semi-online preference optimization that pairs online unpreferred motions with offline preferred examples, addressing offline overfitting and online sampling bias (SoPo, NeurIPS'25).
• Co-developed step-aware reward modeling and reward-guided sampling to improve text-motion alignment and motion realism during denoising (ReAlign, AAAI'26; co-first author).

Southeast University, Nanjing, China — Apr 2024 – Present
Master's Student @ PALM Lab — *Mentor:* Hongsong Wang
• Co-developed step-wise reward optimization and a unified reward model for heterogeneous motion representations (EasyTune, ICLR'26; MotionRFT, TPAMI'26; co-first author on both). On HumanML3D with MLD, EasyTune achieves FID 0.132 at 22.10 GB peak memory, using 15.22 GB less than DRaFT-50.
• Developed frequency-guided diffusion with perturbation training for skeleton-based video anomaly detection, improving robustness to unseen normal motion and emphasizing global-motion reconstruction (FG-Diff, TIP'26).

Shenzhen University, Shenzhen, China — Oct 2022 – Oct 2023
Undergraduate Research Assistant @ Computer Vision Institute — *Mentor:* Can Gao
• Developed multi-scale granular-ball and fuzzy-rough-set methods for detecting heterogeneous outliers (MGBOD, TKDE'25; first student author), and three-way decision-based multi-view co-detection (TWCOD, IJAR'23; first author).

HONORS & AWARDS

Master's Studies

ICML Gold Reviewer	International Conference on Machine Learning	2026
NeurIPS Top Reviewer	Conference on Neural Information Processing Systems	2026
Tencent-SEU Scholarship (top 2.5%)	Southeast University (¥9,000)	2025
First-Class Academic Scholarship (top 10%)	Southeast University (¥12,000)	2025

Undergraduate Studies

Outstanding Graduate Representative	Shenzhen University (Graduation Speech)	2024
Featured in Admissions Guide	Shenzhen University (as Outstanding Student)	2024
Honor Bachelor Degree (top 3%)	Shenzhen University	2024
Outstanding Graduate (top 5%)	Shenzhen University	2024
LiYuan Star (SZU Special Award, top 0.73%)	Shenzhen University (¥20,000)	2022, 2023
Scholarship of Outstanding Innovative Talent	Guangdong Science College Entrance Exam (top 2%, ¥20,000)	2020